

Simon's Problem

Given $f: \{0,1\}^n \rightarrow \{0,1\}^n$ s.t.

- i) f is 2 to 1
- ii) $f(x) = f(x \oplus s)$

Goal: Find s by querying f .

Complexity for classical computers:

Probabilistically:

$\frac{2^n}{2} + 1$ queries in the worst case $\Rightarrow \mathcal{O}(2^n)$

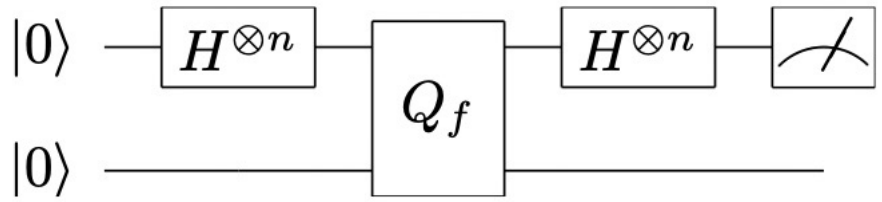
How many queries to observe a collision with high probability?
 $\underbrace{P(x=y) \geq \frac{1}{2}}$

$$P(\text{collision after } t \text{ queries}) = \frac{\binom{t}{2}}{2^n - 1} = \frac{t(t-1)}{2(2^n - 1)} \geq \frac{1}{2}$$

iff $t^2 - t \geq 2^n - 1$

and therefore $t \geq 2^{n/2} \Rightarrow t = \mathcal{O}(\sqrt{2^n})$

Simon's Algorithm



Two steps:

- (1) Run the circuit $m = \mathcal{O}(n)$ times
- (2) Perform classical postprocessing on the output to determine s .

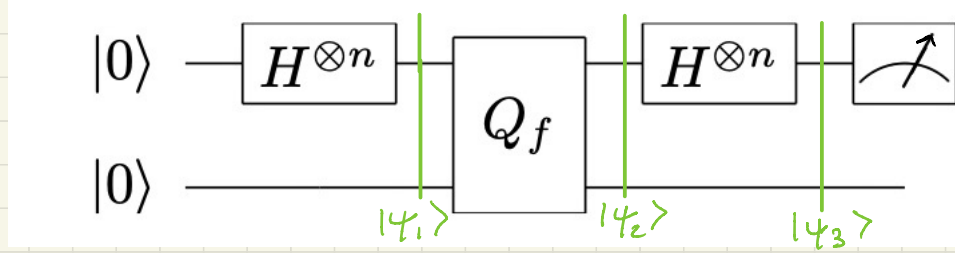
Each run of the circuit allows us to extract a uniformly random string $x_i \in \{0,1\}^n$ such that $\underline{x_i \cdot s = 0}$.
For each of them the amplitude is $\frac{1}{\sqrt{2^{n-1}}}$.

Thus, with $n-1$ runs we obtain $x_1, x_2, \dots, x_{n-1}$ strings and we can calculate s by solving the system of equations $\{x_i \cdot s = 0\}_{i=1, \dots, n-1}$.

Some Remarks

- 1) If $x_1, \dots, x_{n-1}$ are linearly independent, we can solve the system efficiently. No need for n strings as $s = 0^n$ is obviously a solution that we can ignore ($s \neq 0^n$).
- 2) It can be shown that the probability of obtaining $n-1$ lin. ind. strings is greater than $1/2$.
- 3) There is a small chance that we measure the same output string ($x_i = x_j$) at two different runs. Thus m can be not exactly $m-1$.
- 4) The probability that the algorithm fails is bounded (BQP).

Simon's quantum algorithm works on $2n$ qubits as follows:



$$|0\rangle^{\otimes n} |0\rangle^{\otimes n} \xrightarrow{H^{\otimes n}} \underbrace{\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \otimes |x\rangle^{\otimes n}}_{|\psi_1\rangle} \xrightarrow{Q_f} \underbrace{\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle |f(x)\rangle}_{|\psi_2\rangle}$$

Now, suppose that we observe the n -bit string $f(z)$ in the 2nd register. Then, the 1st register will accordingly be in state $\frac{|z\rangle + |z \oplus s\rangle}{\sqrt{2}}$. Therefore, by applying $H^{\otimes n}$ to the 1st register we get:

$$\frac{1}{\sqrt{2}} \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{z \cdot y} |y\rangle + \sum_{y=0}^{2^n-1} (-1)^{(z \oplus s) \cdot y} |y\rangle = \underbrace{\frac{1}{\sqrt{2^{n+1}}} \sum_{y=0}^{2^n-1} (-1)^{z \cdot y} (1 + (-1)^{s \cdot y}) |y\rangle}_{|\psi_3\rangle}$$

Now, the final measurement on the input register will give uniformly random strings s such that $s \cdot z = 0$.

This is because the probability we get for $s \cdot z = 1$ is zero.

Therefore, at each run of the circuit we obtain, with prob.

$$\frac{2}{\sqrt{2} \sqrt{2^{n+1}}} = \frac{1}{\sqrt{2^n}} \quad \text{a } n\text{-bit string } z, \text{ and collecting } n-1$$

strings such that $\{s z_i = 0\}_i$ are linearly independent equations ($z = 0 \dots 0$ is a trivial solution). Solving such a system will give the period s of the function.

We calculate a simple example to show practically how the algorithm works.



Example :

$$f: \{0,1\}^3 \rightarrow \{0,1\}^3$$

$s = ?$

$$\begin{aligned} f(000) &= 010 \\ f(001) &= 101 \\ f(010) &= 101 \\ f(011) &= 010 \end{aligned}$$

$$\begin{aligned} f(100) &= 110 \\ f(101) &= 001 \\ f(110) &= 001 \\ f(111) &= 110 \end{aligned}$$

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} (-1)^{z \cdot x} |z\rangle$$

$$|000000\rangle \xrightarrow{H^{\otimes 3} \otimes I^{\otimes 3}} \frac{1}{\sqrt{8}} \left(|000\rangle |000\rangle + |001\rangle |000\rangle + |010\rangle |000\rangle + |011\rangle |000\rangle + |100\rangle |000\rangle + |101\rangle |000\rangle + |110\rangle |000\rangle + |111\rangle |000\rangle \right)$$

$$\xrightarrow{Q_f} \frac{1}{\sqrt{8}} \left(\cancel{|000\rangle |010\rangle} + \cancel{|001\rangle |101\rangle} + \cancel{|010\rangle |101\rangle} + \cancel{|011\rangle |010\rangle} + \cancel{|100\rangle |110\rangle} + \cancel{|101\rangle |001\rangle} + \cancel{|110\rangle |001\rangle} + |111\rangle |110\rangle \right)$$

$f(x) = 110$:

$$H^{\otimes 3} \left(\frac{1}{\sqrt{2}} (|100\rangle + |111\rangle) \right) = \frac{1}{\sqrt{8}\sqrt{2}} \left(\sum_{z \in \{0,1\}^3} (-1)^{100 \cdot z} |z\rangle + \sum_{z \in \{0,1\}^3} (-1)^{111 \cdot z} |z\rangle \right)$$

Example :

$$f: \{0,1\}^3 \rightarrow \{0,1\}^3$$

$$s = ?$$

$$f(000) = 010$$

$$f(001) = 101$$

$$f(010) = 101$$

$$f(011) = 010$$

$$f(100) = 110$$

$$f(101) = 001$$

$$f(110) = 001$$

$$f(111) = 110$$

$$= \frac{1}{\sqrt{2}\sqrt{8}} \left[\underline{|000\rangle}(1+1) + \cancel{|001\rangle}(1-1) + \cancel{|010\rangle}(1-1) + |011\rangle(1+1) + \right. \\ \left. + |100\rangle(-1-1) + \cancel{|101\rangle}(-1+1) + \cancel{|110\rangle}(-1+1) + |111\rangle(-1-1) \right]$$

$$\left(\frac{2}{\sqrt{2}\sqrt{8}} \right)^2 = p(000) = p(100) = p(111) = p(011) = \frac{1}{4}$$

$$\begin{cases} \cancel{000} \cdot s = 0 \\ 100 \cdot s = 0 \\ 111 \cdot s = 0 \\ 011 \cdot s = 0 \end{cases}$$

$$\Rightarrow s = 011$$